

Protocol Analysis Report

Report type	Network protocol analysis, controlled lab
Version / date	2.0 / 2026-10-07
Capture window (UTC)	2026-10-03 20:38 - 21:00 and 2026-10-05 14:00 - 14:08
Data analyzed	Zeek logs (conn, dns, http, dhcp, ntp, packet_filter) and evidence PCAPs (beacon, dns, port4444)
Environment	One Windows victim host and one Kali Linux analysis host

1. Summary

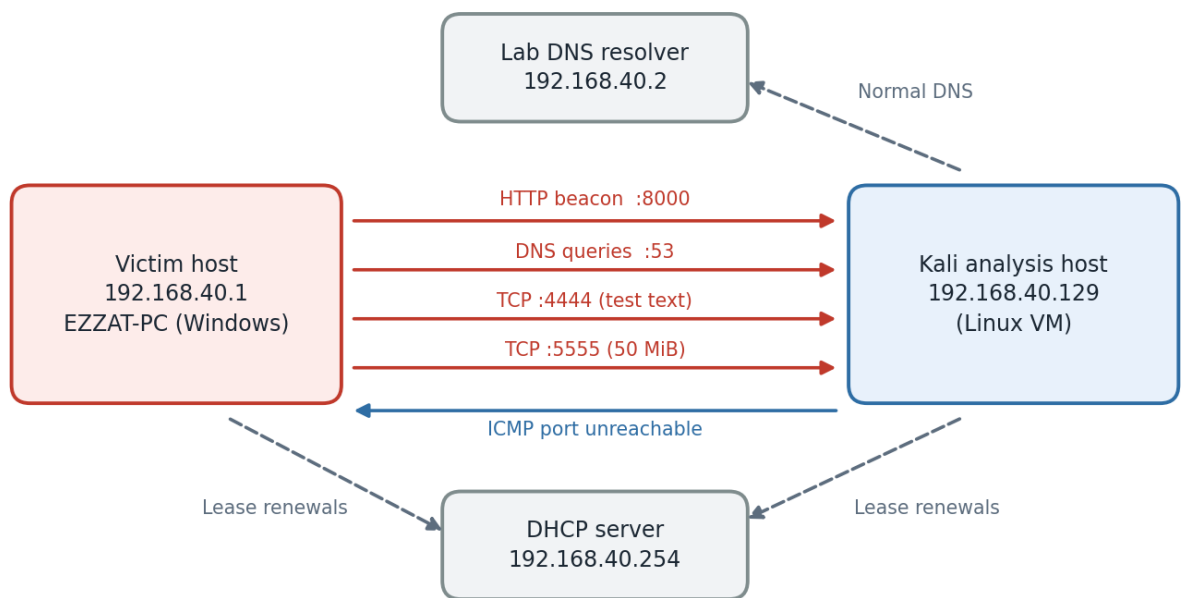
This report explains what each network protocol did in the lab and whether its behavior was normal. Four behaviors were created on purpose to look like attack traffic. All four were started by the victim host toward the Kali host.

165 connections were logged. **99 (60%)** are routine background traffic such as time sync, DNS and browser updates. The other **66 (40%)** belong to the four behaviors below.

#	What happened	Protocol	What it means	Finding
1	30 identical web requests exactly 30 seconds apart	HTTP, TCP/8000	A program checks in on a timer, like malware "beaconing"	F-001
2	33 DNS queries using random-looking names under <code>malicious-c2.xyz</code>	DNS, UDP/53	Looks like hidden data in DNS, but nothing answered	F-002
3	One 50 MiB transfer in about 1 second	TCP/5555	A large file moved to another host	F-003
4	A session on port 4444 that sent the text <code>LAB_TEST</code> three times	TCP/4444	A port often used by attack tools, but the content was harmless	F-004

All four match the lab exercise, so none is a real intrusion. They are described as if they were real, to show how each would be recognized.

2. Network and Hosts



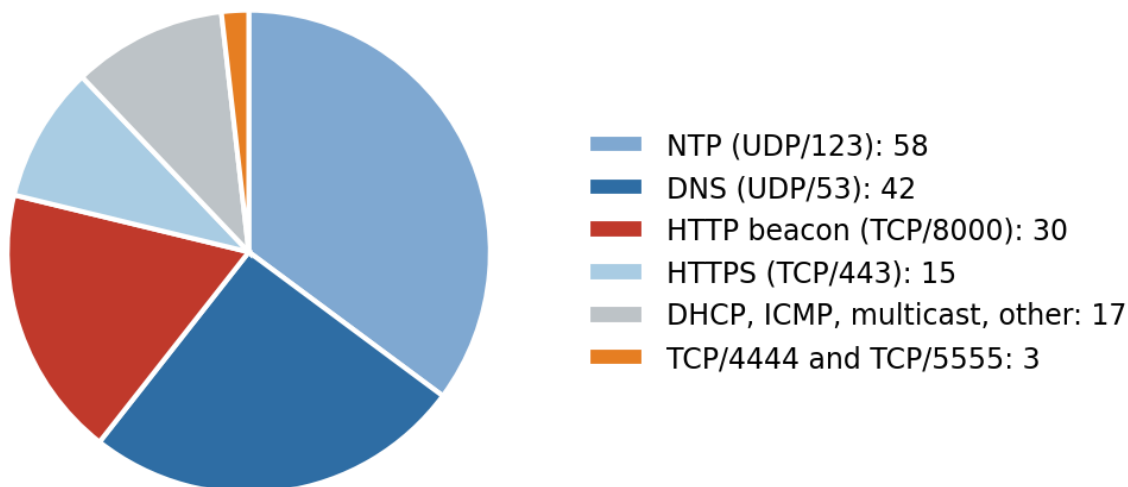
Host	IP address	Hostname	Role
Victim	192.168.40.1	EZZAT-PC	Windows host; starts every unusual flow
Kali analysis host	192.168.40.129	none	Linux VM; receives the unusual flows
DNS resolver	192.168.40.2	none	Answers normal lookups
DHCP server	192.168.40.254	none	Hands out and renews addresses

The roles come from the lab description. The packets agree: the victim's traffic carries a TTL of 128 (Windows default) and the Kali host's carries 64 (Linux default).

3. Traffic Overview

Protocol / port	Connections	Share	What it is	Verdict
UDP/123	58	35.2%	NTP time sync	Normal
UDP/53	42	25.5%	DNS	Mixed
TCP/8000	30	18.2%	HTTP to Kali host	Suspicious
TCP/443	15	9.1%	HTTPS (Mozilla)	Normal
UDP/67	5	3.0%	DHCP	Normal
ICMP / ICMPv6	5	3.0%	Ping, unreachable, multicast reports	Normal
UDP/5355	4	2.4%	LLMNR name queries	Normal
TCP/4444	2	1.2%	Unusual port	Suspicious
IGMP	2	1.2%	Multicast membership	Normal
TCP/5555	1	0.6%	Large transfer	Suspicious
UDP/5353	1	0.6%	mDNS	Normal

Connections by protocol (165 total)



The four behaviors are 66 of the 165 connections (30 beacons, 33 DNS flows to the Kali host, and 3 on ports 4444 and 5555). The single port 5555 flow carries over 99.9% of all bytes sent by clients.

4. Unusual Protocol Behavior

4.1 HTTP beaconing, TCP/8000 (F-001)

In plain terms: a script asks the Kali host for `/beacon` every 30 seconds, like a device repeatedly phoning home.

Detail	Value
Direction	192.168.40.1 → 192.168.40.129:8000
Requests	30, from 2026-10-03 20:45:49 to 21:00:20 UTC
Request	GET /beacon, identical every time
Client	python-requests/2.34.2 (a script, not a browser)
Gap between requests	Mean 30.03 s, range 29.77 – 30.42 s
Server reply	The same "404 File not found" page each time (335 bytes)

Why it stands out: the timing barely varies, the client is a script, and the address never changes. Because the server only answers 404, no commands or data were exchanged.

4.2 DNS, UDP/53 (F-002)

In plain terms: DNS turns names into addresses. Here the victim asked the Kali host directly for random-looking names, a pattern attackers use to hide data inside DNS.

Detail	Value
Time	2026-10-05 14:03:41 – 14:06:03 UTC
Queries	33 sent to 192.168.40.129:53 (not the real resolver)

Detail	Value
Made up of	26 address lookups for 7 names, plus 7 reverse lookups of the Kali host's own address
Pattern	Seven runs, each retrying every 2 seconds, like the nslookup tool
Answers	None. Every query got an ICMP "port unreachable" reply, so nothing was listening

Names queried (each under .malicious-c2.xyz):

Name	Length
random-subdomain	16
lxpht4i9n7vqug5w3mfr	20
t7d2m1i6ycja8hbeg54z	20
f83s5xveji7tn1q2ugbr	20
l7fthbajv4q9p32mcoys	20
m9yjsbv2rk0a734qtngz	20
capf0hr7x65bqn981duw	20

Why it stands out: long random names can carry hidden data. Here each 20-character name uses 20 different characters, which suggests a script made them. Nothing answered, so no data exchange is shown.

4.3 Port 4444, TCP/4444 (F-004)

In plain terms: port 4444 is the default for several hacking tools, so it draws attention. In this session the victim only sent the words LAB_TEST.

Detail	Value
Direction	192.168.40.1 → 192.168.40.129:4444
Started	2026-10-03 20:38:41 UTC (complete handshake)
Length	204.9 seconds, left open
Data	LAB_TEST sent 3 times (27 bytes), at +43 s, +77 s and +205 s
Server data	None

Why it stands out: the port is notable, but the content is plain labelled text with no commands or binary data, and the timing is irregular.

4.4 Large transfer, TCP/5555 (F-003)

In plain terms: one short connection moved exactly 50 MiB from the victim to the Kali host, the size of a typical test file.

Detail	Value
Direction	192.168.40.1:65137 → 192.168.40.129:5555
Time	2026-10-03 about 20:52:59 UTC
Duration	0.99 seconds
Sent / returned	52,428,800 bytes (50 MiB) / 0 bytes

These figures come from Zeek's connection log. Zeek recorded only 56 packets for this flow, so treat the byte total as Zeek's count.

5. Normal Protocols

Protocol	Port	What it does	What was seen
NTP	UDP/123	Keeps clocks accurate	57 polls about every 30 s to three time servers
DHCP	UDP/67	Hands out addresses	5 lease renewals about 900 s apart (1800 s lease)
HTTPS	TCP/443	Encrypted web traffic	15 flows to 199.232.81.91 and 34.107.243.93, matching Mozilla DNS answers
LLMNR / mDNS	UDP/5355, 5353	Windows name lookup on the local network	ezz-at-pc queries at 20:42:17 and 20:57:26 UTC
ICMP	n/a	Network status messages	A ping at 14:02:57 UTC, then 33 "port unreachable" replies to the DNS queries
IGMP / MLD	n/a	Multicast membership	Two reports from the victim host

LLMNR is normal on Windows, but on real networks attackers can abuse it, so many sites turn it off.

6. Timeline

Time (UTC)	Protocol	Event
2026-10-03 20:38:41	TCP/4444	Session to the Kali host begins
2026-10-03 20:45:49	HTTP	First /beacon request
2026-10-03 20:52:59	TCP/5555	50 MiB transfer
2026-10-03 21:00:20	HTTP	30th and last beacon
2026-10-05 14:02:57	ICMP	Ping from victim to Kali host
2026-10-05 14:03:41	DNS	First malicious-c2.xyz run
2026-10-05 14:06:03	DNS	Last captured query

7. Risk Summary

Protocol	Verdict	Severity	Finding
HTTP, TCP/8000	Suspicious, lab-generated	Medium	F-001
DNS, UDP/53	Suspicious, lab-generated	Medium	F-002
TCP/5555	Suspicious, lab-generated	Medium	F-003
TCP/4444	Suspicious, lab-generated	Medium	F-004
NTP, DHCP, HTTPS, LLMNR/mDNS, ICMP, IGMP	Normal	Info	none

8. Recommendations

If this traffic appeared on a real network:

Priority	Action
High	Block or sinkhole <code>malicious-c2.xyz</code> , and allow DNS only to approved resolvers
High	Alert on fixed-interval requests to the same address, especially from script clients like <code>python-requests</code>
High	Review large transfers to uncommon ports and confirm what was sent
Medium	Alert on connections to port 4444 and identify the program on both ends
Medium	Detect long, random-looking DNS names and repeated failed lookups
Medium	Turn off LLMNR where it is not needed

9. Limitations

- Zeek often saw only one direction of a flow, so some counts and byte totals are partial.
- Host roles come from the lab description and TTL values; no host forensics were done.
- HTTPS is encrypted, so only addresses and counts were reviewed.

Appendix — Glossary

Term	Meaning
Beaconing	A program contacting a controller on a regular timer
C2 (command and control)	The channel an attacker uses to direct a compromised host
DNS tunneling	Hiding data inside DNS queries and replies
TTL	A packet counter that differs by operating system (128 Windows, 64 Linux)
Zeek	A network monitor that writes logs about connections and protocols